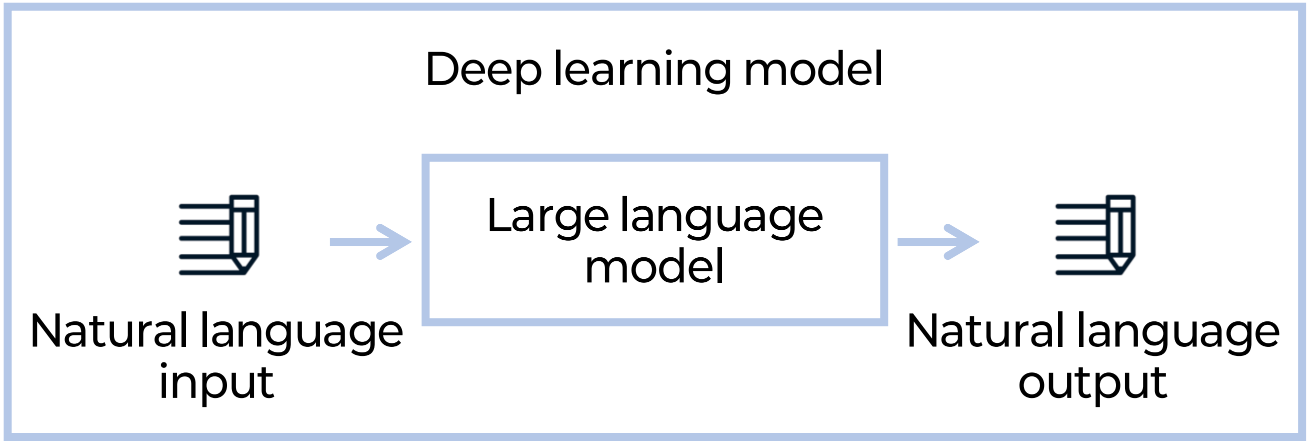
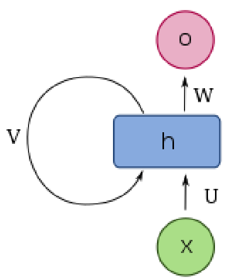
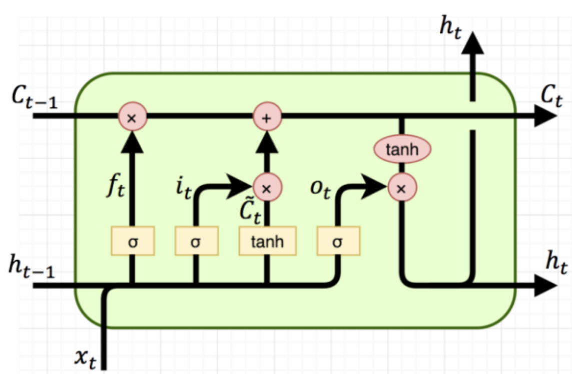
LLM?

Deep learning model trained by large language dataset (corpora)

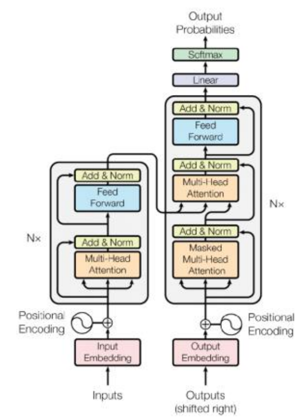




RNN [1]
(Recurrent Neural Network)



LSTM [1]
(Long short-term memory)



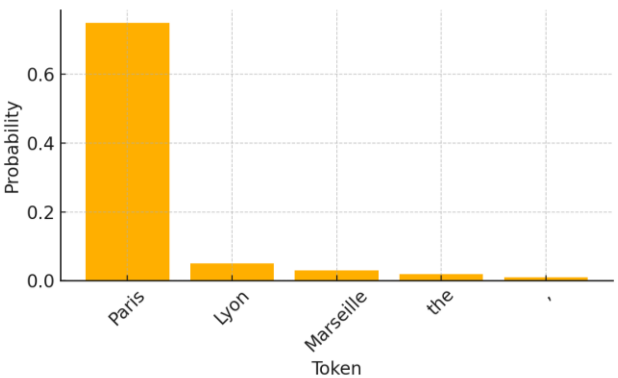
Transformer [2]

[1] Pedro, *Medium*, **2018**
[2] Vaswani et al., *Adv. Neural Inf. Process. Syst.*, **2017**, 30

GPT: Generative Pre-trained Transformer

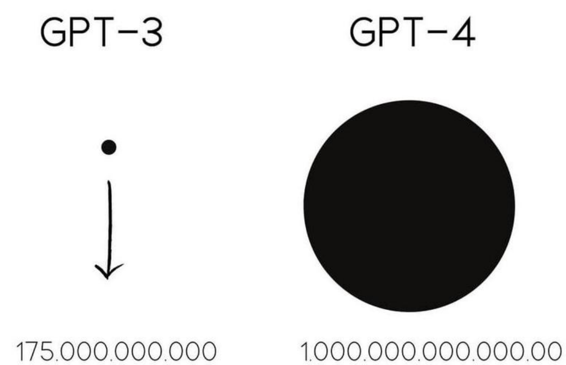
1. Generative

The capital city of France is ________.



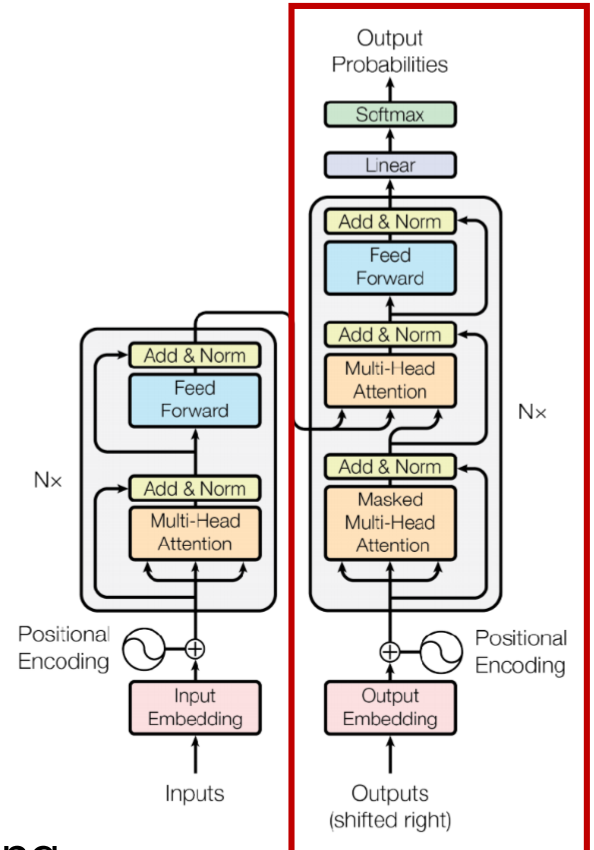
2. Pre-trained [1,2]

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}
GPT-3 Small	125M	12	768	12	64
GPT-3 Medium	350M	24	1024	16	64
GPT-3 Large	760M	24	1536	16	96
GPT-3 XL	1.3B	24	2048	24	128
GPT-3 2.7B	2.7B	32	2560	32	80
GPT-3 6.7B	6.7B	32	4096	32	128
GPT-3 13B	13.0B	40	5140	40	128
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128



$40 million for pre-training
→ individually, impossible

3. Transformer [3]



Decoder-only transformer

[1] Brown et al., *Adv. Neural Inf. Process. Syst.*, **2020**, *33*, 1877. [3] Vaswani et al., *Adv. Neural Inf. Process. Syst.*, **2017**, 30
[2] Omoruyi, *Technext.*, **2025**,

LLM as a mapping

Process of generation

$\mathbf{X}$
NL
Sequence

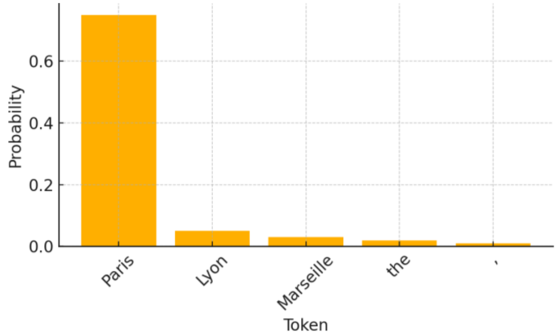
LLM

$p_\theta(\cdot\,|\mathbf{X})$
Probability on
Language space

$f_\theta(\mathbf{X})\colon\ \mathbf{X} \rightarrow p_\theta(\cdot\,|\mathbf{X})$

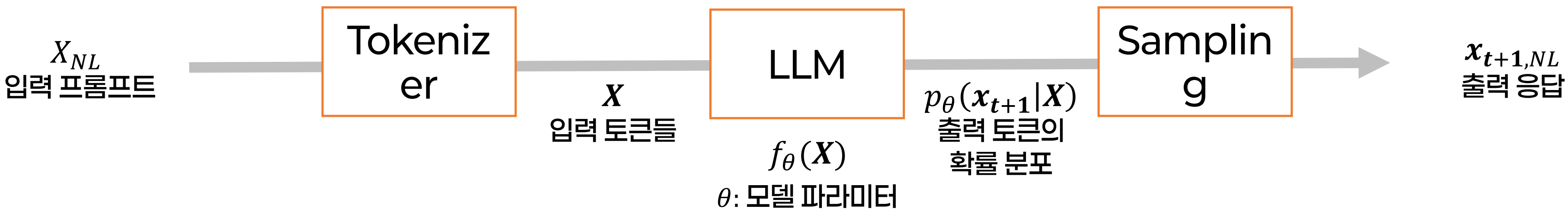
θ: model parameter

The capital city of France is ________.

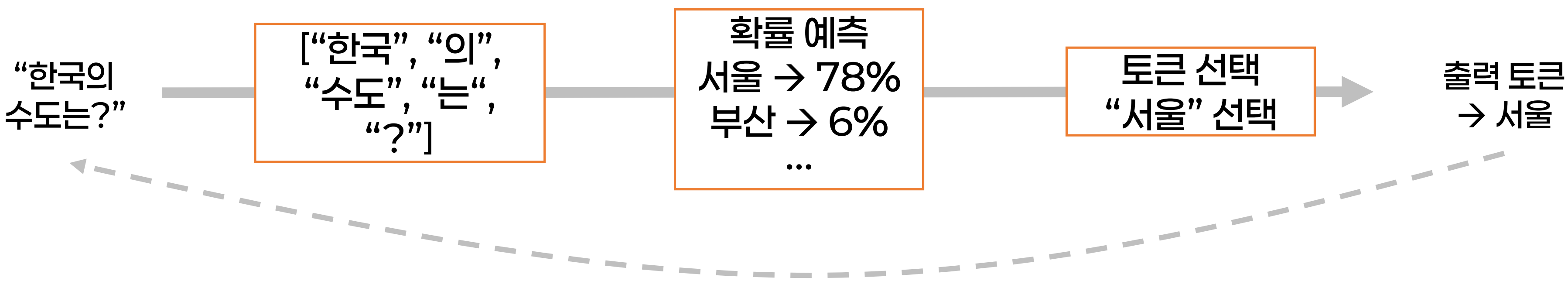


LLM as a mapping

Process of generation

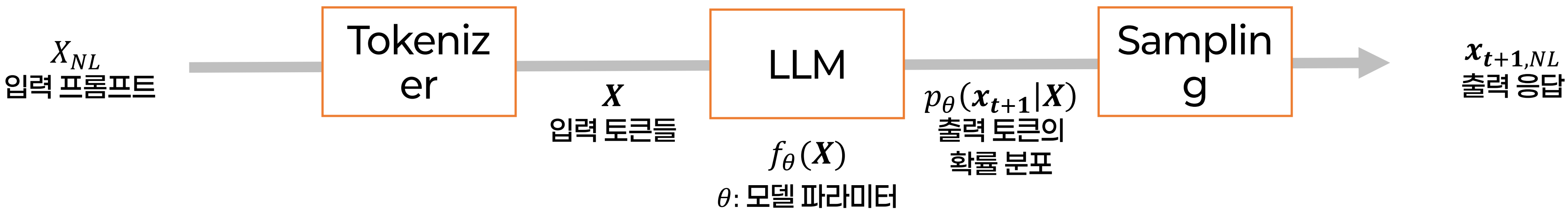


예시)



LLM as a mapping

Process of generation



$$s \xrightarrow{\tau} \mathbf{X} \xrightarrow{\mathbf{E}_\theta} \mathbf{H}^{(0)} \xrightarrow{\Phi_\theta} \mathbf{H} \xrightarrow{W_U,\,\mathrm{softmax}} f_\theta(\mathbf{X}) \xrightarrow{\mathrm{dec}} x_{n+1}. \tag{1}$$

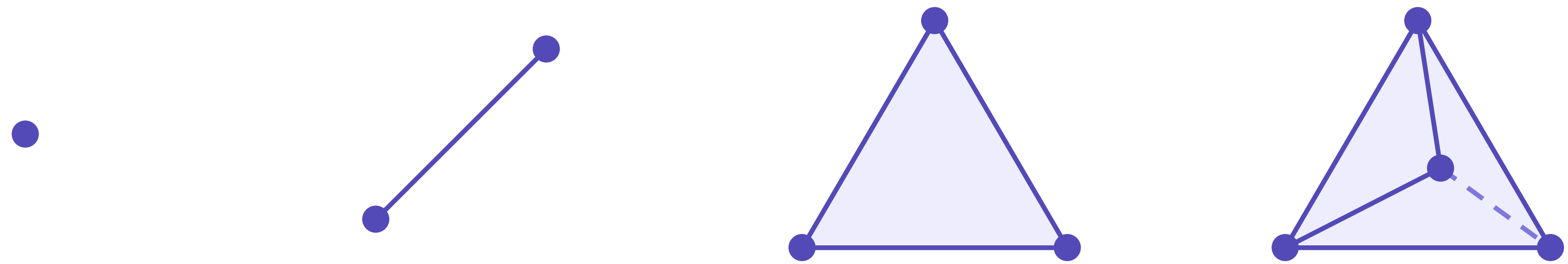
LLM $$f_\theta(\mathbf{X}) \equiv p_\theta(\cdot\,|\,\mathbf{X}) = \mathrm{softmax}(W_U \mathbf{H}_n) \in \Delta^{|\mathcal{V}|-1}$$

VLM $$f_\theta(\mathbf{X}, \mathbf{I}) = \mathrm{softmax}\Big(W_U\,\big[\Phi_\theta([\,P(E_v(\mathbf{I})); \mathbf{E}_\theta(\mathbf{X})\,])\big]_n\Big) \in \Delta^{|\mathcal{V}|-1}$$

$$x_{n+1} \sim f_\theta(\mathbf{X}), \quad \mathbf{X} \leftarrow \mathbf{X} \oplus (x_{n+1}), \quad \text{repeat until } x_{n+1} = \langle \mathrm{EOS} \rangle.$$

Mapping to probability simplex

Simplex?

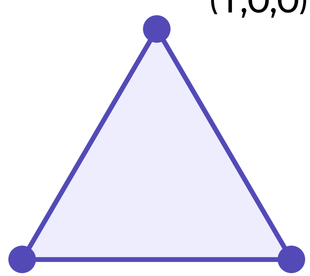


0-simplex
점 · 꼭짓점 1개

1-simplex
선분 · 꼭짓점 2개

2-simplex
삼각형 · 꼭짓점 3개

3-simplex
사면체 · 꼭짓점 4개



2-simplex
삼각형 · 꼭짓점 3개

For LLM,

N-simplex
→ N = number of token
→ Cardinality of the simplex

Model	N
GPT-4o	~200,000
Llama 3	128,256
Qwen 2.5	~150,000

Mapping to probability simplex

Simplex?

$\boldsymbol{X}(\boldsymbol{x}_{1:t})$
자연어
시퀀스
(tokenized)

LLM

$p_\theta(\boldsymbol{x}_{t+1}|\boldsymbol{X})$
언어 벡터 공간
위의 확률분포

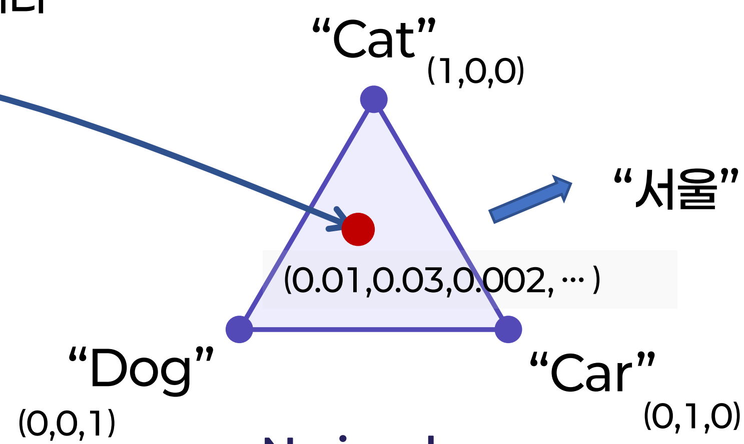
$f_\theta(\boldsymbol{X})$

θ: 모델 파라미터

"한국의
수도는?"

["한국",
"의", "수도",
"는", "?"]

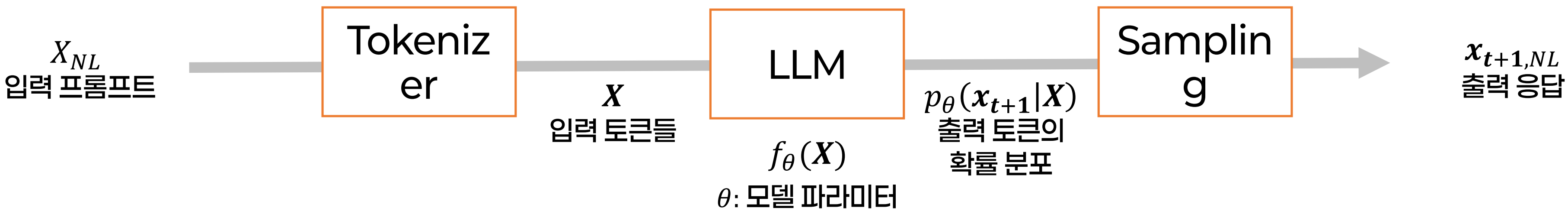
$\boldsymbol{X}(\boldsymbol{x}_{1:t})$ =
[63, 262,
473,
2999, 2]



N-simplex
삼각형 · 꼭짓점 3개

Strategy of designing LLM

3-strategy



전략 1. θ adjustment / $f_{\theta_{opt}}(\mathbf{X})$

→ Model selection, Supervised fine-tuning (SFT), Reinforcement learning (RLHF)

전략 2. X adjustment / $f_\theta(\mathbf{X}_{\text{opt}})$

→ In-context learning (ICL), prompt optimization

전략 3. Arrangement of multiple LLMs, tools, and retrieveres.

/ $p(\boldsymbol{x}_{t+1}|\boldsymbol{X}) \sim f_{\theta_k}(f_{\theta_i}(\boldsymbol{X}) + f_{\theta_j}(\boldsymbol{X}) + Tool_i(\boldsymbol{X}) + Retriever_i(\boldsymbol{X}))$

→ multi-agentic system (MAS), MAS optimization, Harness engineering

Strategy of designing LLM

3-strategy

$$s \xrightarrow{\tau} \mathbf{X} \xrightarrow{\mathrm{E}_\theta} \mathbf{H}^{(0)} \xrightarrow{\Phi_\theta} \mathbf{H} \xrightarrow{W_U,\,\mathrm{softmax}} f_\theta(\mathbf{X}) \xrightarrow{\mathrm{dec}} x_{n+1}. \quad (1)$$

Strategy 1. parameter adjustment $f_{\theta_{\mathrm{opt}}}(\mathbf{X})$

Strategy 2. prompt optimization $\mathbf{X}_{\mathrm{opt}} = \arg\max_{\mathbf{X}\in\mathcal{P}} J(f_\theta(\mathbf{X}))$

Strategy 3. multi-agent $f_{\theta_{\mathrm{out}}}\Big(\underbrace{\mathbf{Y}_{\theta_1}(\mathbf{X}_1)}_{\text{agent}} \oplus \underbrace{T(\mathbf{X}_T)}_{\text{tool}} \oplus \underbrace{R(\mathbf{X}_R;\eta)}_{\text{retriever}} \oplus \cdots\Big) \in \Delta^{|\mathcal{V}|-1}$

Strategy 1+2+3: self-evolving, harness engineering

Practice: RAG

Before lunch

Practice: Multi-agentic system

After lunch